1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

ANS:-

The ping command sends packets to the destination server and measures the response time.

- Website tested: www.youtube.com
- Packets sent: 4 (by default on Windows)
- Average response time: Record the value displayed as Average in your command prompt.
- If the request times out, the destination may be unreachable or may not respond to ICMP packets.

Result: The ping command helps check network connectivity and measure response time.

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Hint: Each hop represents a network device your data passes through to reach the destination.

ANS:-

The tracert command displays the intermediate network hops between your computer and the destination.

Each hop represents a router or network device encountered along the route.

- Hop 1: Usually the local router or gateway.
- Intermediate hops: Routers operated by your ISP or other networks.
- Final hop: The destination, if reached and responsive.

Result: The command helps identify where delays or routing problems may occur.

Note: The number of hops and IP addresses depend on your network and the route at the time of testing. Copy the actual hop list from your terminal.

3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

ANS:-

The nslookup command queries a DNS server to find the IP address associated with a domain name.

- Domain name: www.zomato.com
- DNS server: Record the server shown in your output.
- IP address: Record the address returned under Addresses.

Result: DNS translates a human-readable domain name into an IP address that computers use to communicate.

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

ANS:-

The ping command tests connectivity between my computer and another device on the local network. Here, 192.168.1.5 is an example IP address that must be replaced with the actual address of my friend's laptop or phone.

- If the device is online and allows ping requests, replies are received.
- If the device is offline or unreachable, the command may display Request timed out or Destination host unreachable.
- A timeout can also occur if a firewall blocks ICMP packets.

Result: Ping helps check local network connectivity and identify possible communication problems.

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Constraint: Give a specific example for each tool based on your own test results above.

ANS:-

Report: Using Network Troubleshooting Tools

1. Ping is useful for checking connectivity and response time; for example, ping www.youtube.com shows whether replies are received and how long they take.
2. If a website is not loading, ping can help determine whether the destination responds to network requests.
3. Traceroute (tracert) helps investigate slow connections by showing the route and response times of intermediate hops.
4. For example, tracert www.flipkart.com can help identify a hop where delays or timeouts occur.
5. Nslookup is useful when a DNS problem is suspected because it checks whether a domain name resolves to an IP address.
6. For example, nslookup www.zomato.com displays the DNS server used and the IP address returned.
7. Together, these three tools help diagnose connectivity, routing, and DNS resolution problems.

Important: Run each command on your own computer and replace the example results with your actual observations before submitting the assignment.